HVDC Design

AmWatt Team Presentation

Last Update: September 30, 2026

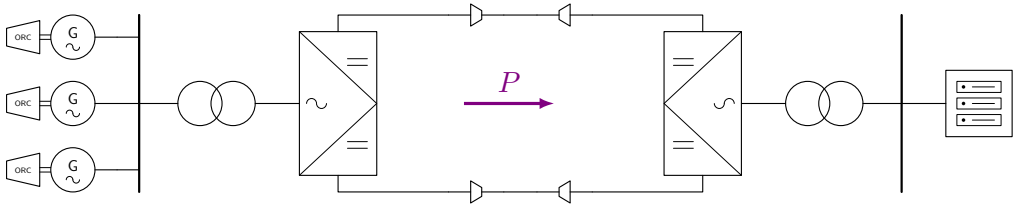


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System Overview

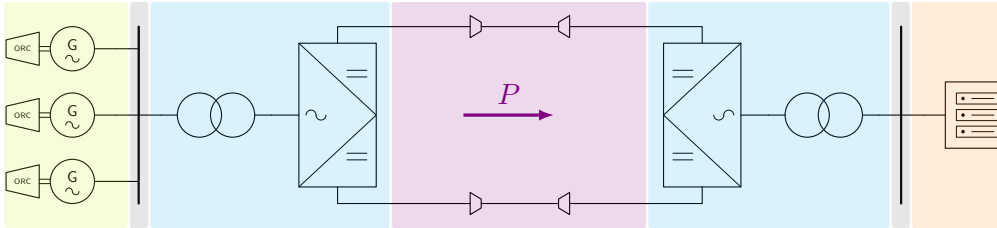
ORC generators feed a data centre through a point-to-point VSC HVDC link.



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System Overview

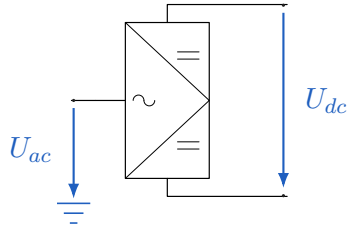
The complete system consists of different elements, supplied by different companies. AmWatt is responsible for the integration of those elements.



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VSC converter

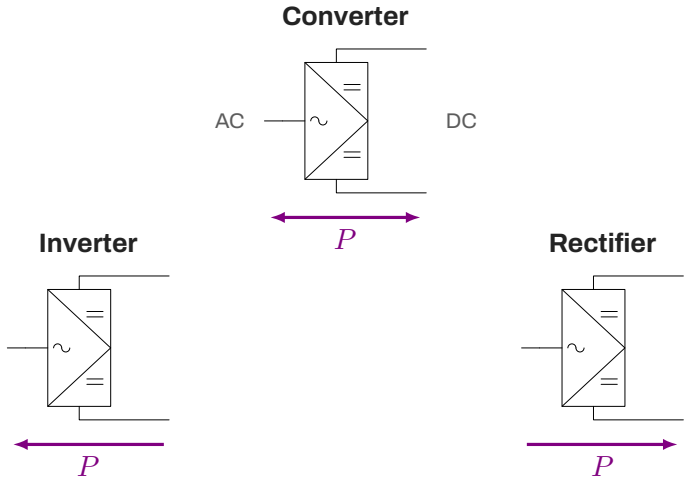
The VSC converts the a.c. voltage U_{ac} into the pole-to-pole d.c. voltage U_{dc}.



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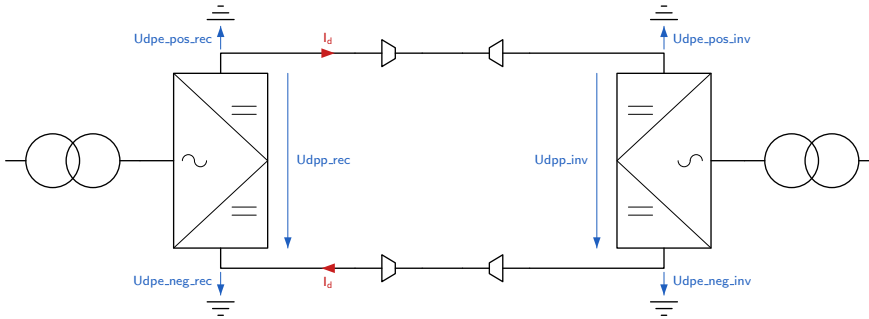
Converter operating modes

The same converter works as rectifier or inverter, set by the direction of P.



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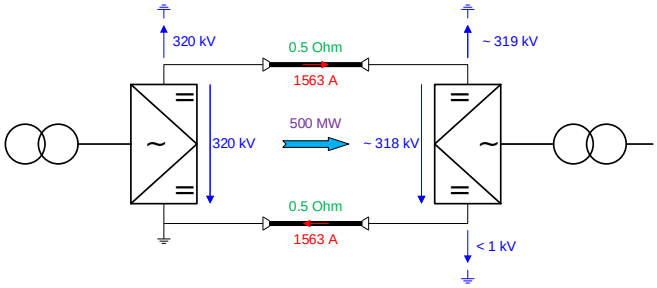
DC load flow definitions



- DC equipment (incl. DC line) voltage is defined by U_{dpe_rec}
- Converter size is defined by U_{dpp_rec}

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Monopole



DC load flow

$$U_{dpp_rec} = U_{dpe_rec}$$

$$I_{DC} = \frac{U_{dpp_rec} - U_{dpp_inv}}{2 \cdot R_{DC}}$$

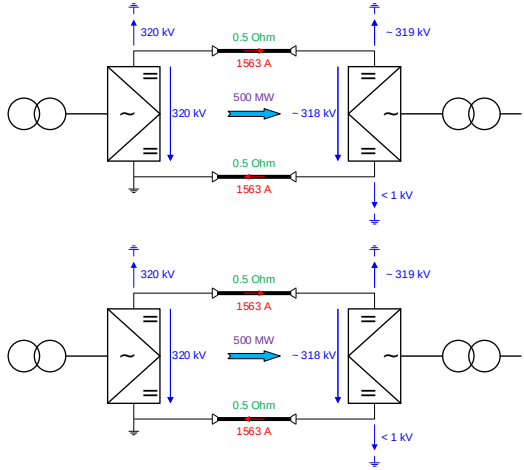
$$U_{dne_rec} = I_d \cdot R_{DC}$$

- Max. voltage created by converter: U_{dpp_rec}
- Max. equipment voltage:
- DC cables have different voltage ratings

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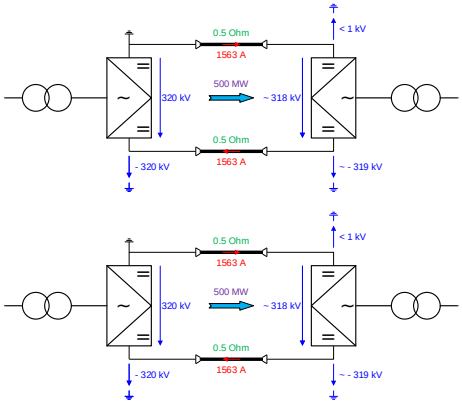
Monopole - extending power transmission

- **Higher DC voltage**
  - Limited equipment availability
  - No redundancy
- **Higher DC current**
  - Higher DC line costs
  - No redundancy
- **Additional monopole**
  - Additional converter equipment
  - 4 DC conductors
  - Increased reliability

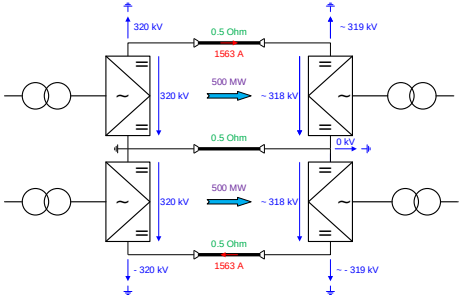


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Monopole → Bipole (BIP)



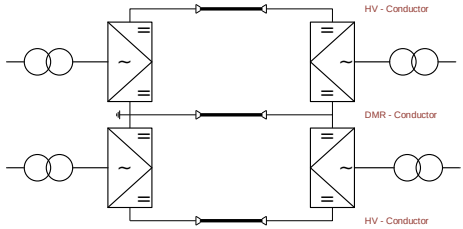
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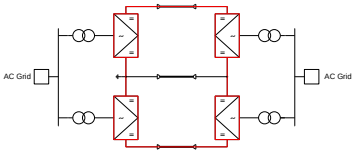
Bipole (BIP)

- 3 conductors of 2 different types.
- Two poles that can transmit power independently.
- On a fault in one pole, the other pole still transmits power — redundancy.
- In symmetrical operation (both poles equal), no current flows through the DMR.



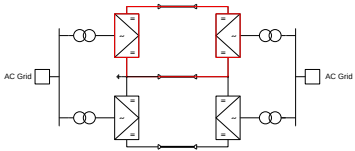
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BIP operation modes



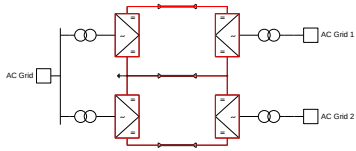
Bipolar — balanced

- Equal power for poles
- No DMR current
- Loss-optimized



Monopolar

- Emergency mode
- System-design definition

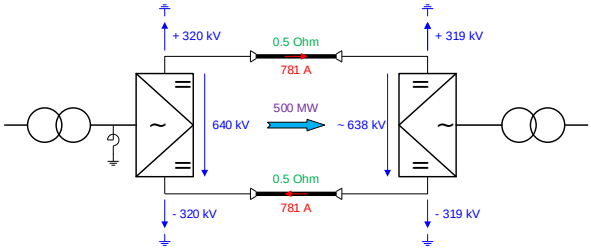


Bipolar — unbalanced

- Split-busbar operation
- DMR current flow

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Symmetrical Monopole (SMP)



- High-impedance earth reference on the AC side
- $U_{dpp_Rec} = 2 \cdot |U_{dpe_Rec}|$
- $I_d = \frac{U_{dpp_rec} - U_{dpp_inv}}{2 \cdot R_d}$
- Only for VSC-HVDC systems
- DC cables have the same voltage rating

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Modular Multilevel Converter (MMC)

Power transmission parameters:

- Power: 1000 MW
- DC voltage: $\pm$ 320 kV
- Max. DC current: 1563 A

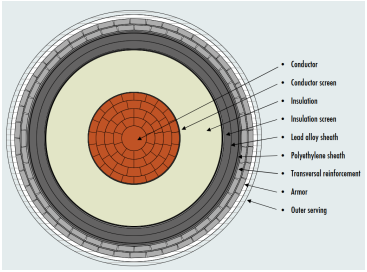
	2 $\times$ Monopole	**1 $\times$ BIP**	**1 $\times$ SMP**
Transformers	$--$	$--$	$++$
DC lines	4 lines / 2 types	3 lines / 2 types	2 lines / 1 type
Converters	4 converters	4 converters	2 converters
Converter rating (U_{dpp})	320 kV	320 kV	640 kV
Losses	$--$	$-$	$+$
Reliability	$++$	$+$	$--$

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DC cable

- **Insulator** — XLPE (cross-linked polyethylene) or MI (mass-impregnated).
- **Conductor** — copper or aluminium.
- Balance between DC voltage and DC current for the most economical design.
- Insulator is generally cheaper than conductor $\rightarrow$ higher voltage rating is preferred

Project voltages: 80/200/250/**320**/400/**525** kV (525 kV is new standard for ≥ 2000 MW).





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Topology comparison

	Sym. Monopole	Rigid Bipole	Bipole w/ MR
DC conductors	2	2	3
Power on converter fault	0%	50%	50%
Power on DC conductor fault	0%	0%	50%
Economical DC voltage	≤ 400 kV	> 500 kV	> 500 kV
Costs	$	$$	$$$
Max. power	1400 MW (320 kV) 1500 MW (400 kV)	2000 MW	3000 MW

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